SIEMENS

INSTALLATION/WIRING INSTRUCTIONS
MODELS SA-3I AND SA-11P
AIR DUCT DETECTORS

INTRODUCTION

Model SA-3I and SA-11P detectors from Siemens Industry, Inc. are self-contained air duct detectors designed for installation in heating, ventilating, and air conditioning duct systems. These Detector Models SA-3I and SA-11P can use ionization or photoelectric smoke detectors to signal the presence of combustion products in a duct system. (Refer to **TECHNICAL DATA** for information on compatible detectors.)

In addition to the self-contained detection features of the SA-3I and SA-11P, these air duct detectors also can power and supervise the following configurations.

(1) two air duct detectors (Model AD-3I or AD-11P) mounted on the exterior of the duct and employing sampling tubes to monitor the air in the duct, **or**

(2) two spot detectors (Model PE-11) mounted on the interior wall of the air duct, **or**

(3) one each of the detectors listed in (1) and (2) above.

TECHNICAL DATA

Smoke Detector Compatibility:	SA-3I - Use Model DI-B3 or DI-B3H only SA-11P - Use Model PE-11 AD-3I (Satellite) - Use Model DI-B3 or DI-B3H only AD-11P (Satellite) - Use Model PE-11
Operation Voltage and Current:	120 VAC (50/60 Hz), .32A max
Alarm Relay:	DPDT contacts, rated at 125 VAC/24 VDC, 3A Resistive
Trouble Relay:	SPDT contacts, rated at 125 VAC/24 VDC, 2A Resistive
Smoke Detector Sensitivity:	Factory set for air duct application—See detector nameplate label
Alarm Indicator:	LED on smoke detector displays steady or flashing red light through air duct cover
Temperature Range:	32°F (0°C) to 100° F (37.8^{O}C) per UL 268/268A
Altitude Range:	SA-3I, using DI-B3 - Altitude range is 0 to 4000 feet above sea level SA-3I, using DI-B3H - Altitude range is 3000 to 8000 feet above sea level SA-11P, using PE-11 - No altitude limitations; detectors unaffected by altitude
Relative Humidity Range	0 to 93% (non-condensating) RH per UL 268/268A
Air Duct Velocity Range:	SA-3I: 500 to 4000 ft/min SA-11P: 300 to 4000 ft/min
Sampling Tube Pressure Differential Range:	SA-3I: 0.017 to 1.07 inches of water SA-11P: 0.006 to 1.07 inches of water

CAUTION: These air duct detectors are designed for detection of products of combustion in a duct system. They are not to be used for open area protection.

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OPERATION

When the SA-3I *or* SA-11P is in operation, a sample of air is drawn from the duct and passed through the sampling chamber at low velocity by means of the inlet sampling tube. The air sample passes through the smoke detector mounted in the duct housing and is exhausted back into the duct via the outlet sampling tube.

Standby Indication

During normal operation the green Power LED on the power supply control board, located in the wiring compartment of the SA-3I/-11P, is lit, indicating that AC power is being applied. The alarm and trouble relays are in non-alarm and non-trouble states during standby operation.

Alarm Indication

Each smoke detector used with the SA-3I/-11P includes an LED for visual alarm indication. If the air duct detector is not visible, a remote alarm lamp can be used for visual alarm indication. (See Figure 6 for models.)

Alarm Operation

An alarm signal from the smoke detector within the SA-3I/-11P or from a satellite detector causes the alarm relay in the SA-3I/-11P to energize.

The Model SA-3I/-11P can be wired for direct alarm relay control of normally energized and de-energized devices. Refer to **ELECTRICAL WIRING** for details. When all products of combustion are cleared from the duct system, the Model SA-3I/-11P can be reset by operating the **RESET** switch with the Reset key.

WARNING: Air duct detectors SA-3I/-11P **cannot** be used for releasing service.

Trouble Condition

A trouble condition in the SA-3I/-11P or in a satellite unit due to a loss of detector supervision causes the trouble relay in the SA-3I/-11P to de-energize. Loss of AC power to the SA-3I/-11P results in the green Power LED turning off and the trouble relay de-energizing.

MOUNTING THE AIR DUCT HOUSING

Location on Duct System

The air duct detector should be located in the main supply duct, downstream from the filters and positioned so as to operate reliably in case of smoke in any part of the air stream. In instances where the filters are capable of removing smoke, a detector should be installed both upstream and downstream from the filters.

The air duct detectors, because they use sampling tubes which monitor the full width of an air duct, overcome the limitations of spot-type smoke and heat detectors in the duct. However, since stratification can occur in the air stream after a long duct run, it is desirable to locate the detector after bends or inlets which create turbulence, and hence, a more homogeneous mixture of air. The detector should, when possible, be located a minimum of six duct-widths downstream from the source of turbulence (see Figure 1). A 12 inch by 12 inch access hole should be cut in the duct adjacent to the detector to permit checking and cleaning the sampling tubes.

The air duct detector should be located in the air handling system, as shown in Figure 2, and should be in conformance with NFPA Pamphlet No. 90A, **Air Conditioning and Ventilating Systems** and with NFPA 72 **National Fire Alarm Code**. (Both publications are available from the National Fire Protection Association, Batterymarch Park, Quincy, Massachusetts.) The detector on the return air side of the blower should be located at a point prior to exhausting air from the building or diluting return air with outside air. The detector on the supply air side of the blower should be downstream of the blower.

The detectors should be wired into the system so that they automatically shut down the blowers and operate dampers as required.

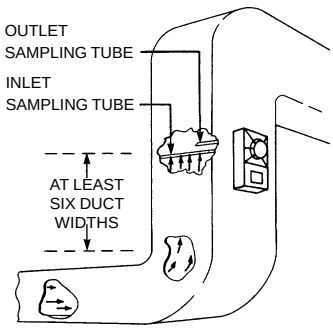
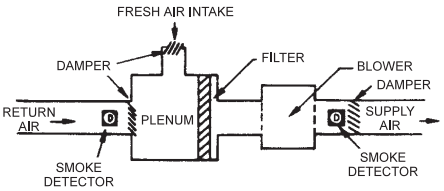


Figure 1
Typical Mounting of Duct

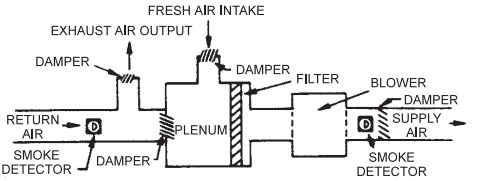
Mounting the Air Duct Housing (See Figure 3)

a. Affix the adhesive backed gaskets (Item 1) to the back of the detector housing (2) so that the larger hole on the gasket lines up with the sampling tube hole on the sampling chamber and the smaller hole with the housing mounting hole.

TYPE A: CLOSED SYSTEM - NO EXHAUST



TYPE B: PROVISION FOR EXHAUSTING SOME PERCENTAGE OF RETURN AIR



TYPE C: RETURN AIR UNDER POSITIVE PRESSURE FROM SEPARATE BLOWER

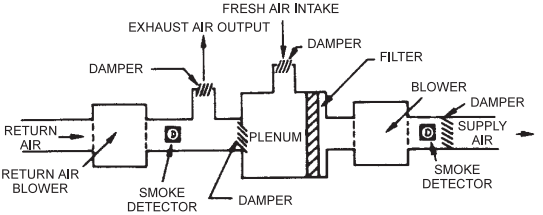


Figure 2
Recommended Locations in Duct Systems

ITEM No.	COMPONENT	ITEM No.	COMPONENT	ITEM No.	COMPONENT	ITEM No.	COMPONENT
1	(2) Sampling Tube Gasket	7	(2) Screw, No. 10 - 1	13	(1) Plug	19	(1) DB-ADPT Base Adapter
2	(1) Housing	8	(2) Lockwasher	14	(1) Detector PE-11	20	(1) Ionization Detector Cover
3	(1) Inlet Sampling Tube	9	(1) Rubber Stopper	15	(1) Detector DI-B3, DI-B3H	21	(1) Sensitivity Test Jack Lid
4	(1) Outlet Sampling Tube	10	(3) Bracket	16	(1) Cover Gasket		
5	(1) Template	11	(3) Screw, 3/16	17	(1) PE-11 Detector Cover		
6	(2) Screw, No. 10 - 1 1/2	12	(3) Screw, 8/32	18	(8) Screw, No. 6		

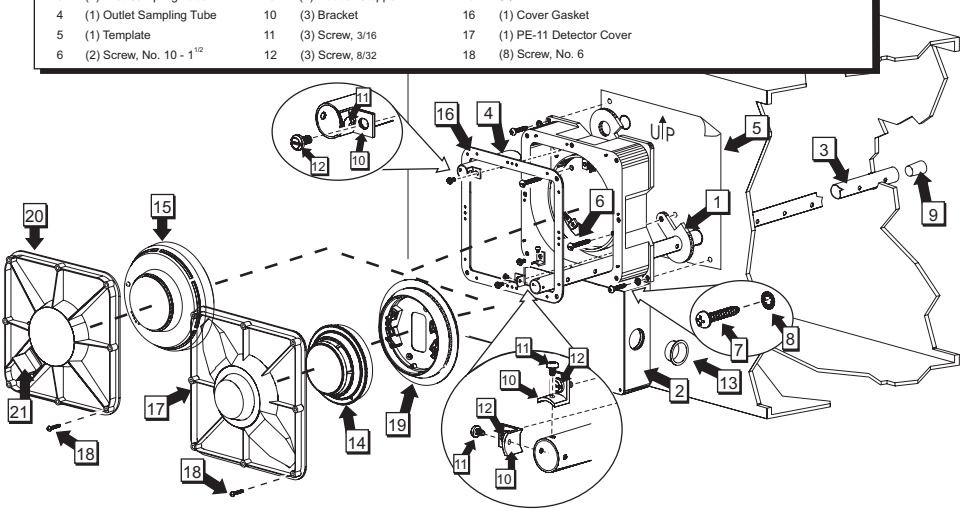


Figure 3
Exploded View

b. Affix the adhesive backed template (5) to the side of the ductwork.

c. Drill the four 0.110 diameter holes with a No. 35 drill bit and cut or punch out two 7/8-inch diameter holes as indicated on the template.

d. Attach the appropriate model housing to the air duct using two No. 10 1½ inch screws (6) and also using two No. 10 1-inch screws (7) with lockwashers (8).

e. Cut a 12 inch by 12 inch access hole in the duct adjacent to the housing to permit checking and cleaning of the sampling tubes. The access hole must be covered and sealed when not in use.

Sampling Tube Selection

To insure accurate air sampling, inlet sampling tubes are available in four lengths. To select the appropriate length, determine the outside width of the duct. Select the sampling tube nearest to, but greater than, the duct width (Table 1). The outlet sampling tube for all ducts is a fixed length of 4 inches (10.1 cm). Trim the inlet sampling tube at the job site as described below.

TABLE 1 - Sampling Tube Selection Chart

Outside Duct Width	Sampling Tube Model No.	Standard Tube Length
1 ft 9 ¾ in or less	STA-2	2 ft
1 ft 9 ¾ in - 3 ft 3 ¾ in	STA-3	3 ft 6 in
3 ft 3 ¾ in - 6 ft 3 ¾ in or less	STA-6	6 ft 6 in
6 ft 3 ¾ in - 9 ft 9 ¾ in or less	STA-10	10 ft
Greater than 9 ft 9 ¾ in	Consult factory	

Note: Each model is manufactured with a different number and size of hole. Therefore, only the specific tube must be used for the specified duct width.

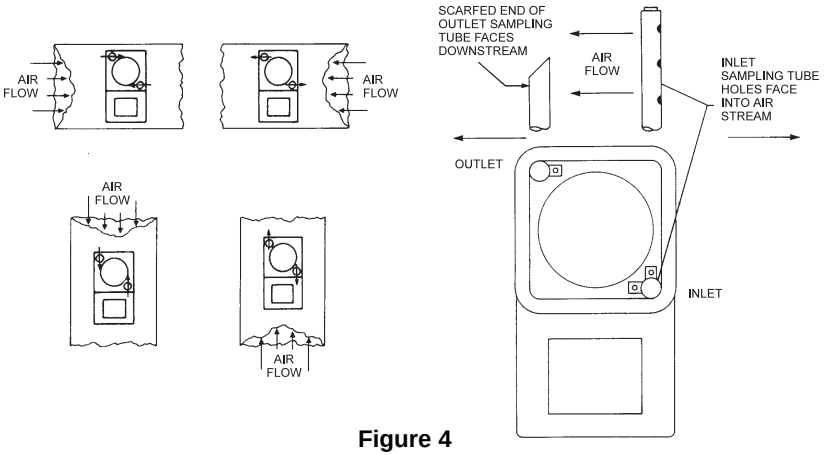


Figure 4
Sampling Tube Orientation

Trimming the Inlet Sampling Tube (See Figure 4)

a. Measure the outside width of the duct.

b. Add 2.25 inches to obtain proper length of tube.

Example:

Outside width of duct		2 ft 6in
Add 2.25 inches	+	2.25 in
Length of sampling tube		2 ft 8.25 in

The length (2 ft 8.25 in) determines (Table 1 above) the model number of the appropriate sampling tube (STA-3).

c. Measure the sampling tube, starting from the end with the four mounting holes.

d. Trim tube to the correct length, remove all burrs, and insert a rubber stopper (Figure 3, Item 9) in the tube.

Installation of the Sampling Tubes

a. Before installing the tubes, cut a 7/8-inch hole in the duct wall that is directly opposite the inlet sampling chamber hole in the detector housing. This will allow the inlet tube to protrude through the duct.

b. Attach brackets (10) to the inlet and outlet air sampling tubes with 3/16 screws (11). **NOTE: When attaching brackets, be sure to align tubes properly (See Figure 4).**

c. Install the inlet tube by inserting it in through the air sampling chamber as shown in Figure 3 (3). Face sampling tube holes into the air stream (Figure 4), and secure to the housing with 8/32 screws (12).

d. Install the outlet tube (4) by inserting it into the upper socket of the air sampling chamber with the scarfed end facing downstream (See Figure 4**).** Secure the bracket to the housing with an 8/32 screw.

e. Make sure no sampling holes extend beyond the ductway, that the stopper remains firmly in position, and that the duct hole is sealed around the protruding sampling tube.

Air Duct Sampling Tube Pressure Measurement

The Model PDM-3 Pressure Differential Measuring device should be used to ensure that the sampling tube pressure differential is within the specified limits (See **TECHNICAL DATA).** Measurements should be taken by qualified personnel in accordance with the **PDM-3 Instructions,** P/N 315-085535.

ELECTRICAL WIRING

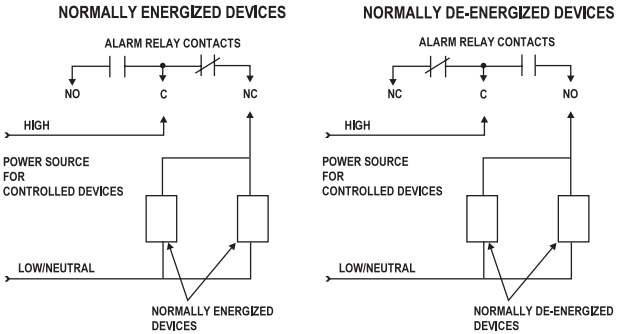
The Air Duct Detector Model SA-3I/-11P is interconnected and wired following the wiring diagram in Figure 6. Each model requires a two-wire circuit of 18 AWG min thermoplastic fixture wire, enclosed in conduit, or of 18 AWG min limited energy shielded cable, without conduit, if permitted by local building codes.

Control of Normally Energized and De-Energized Devices

The Model SA-3I/-11P can be wired for direct relay control of normally energized and deenergized devices as follows (Refer to Figure 5).

a. Wire normally energized devices (fans, blowers, etc.) to the NC contacts between terminals 10 and 11 or 13 and 14 in order to shut down these devices on alarm.

b. Wire normally de-energized devices (extinguishing smoke removal fan, etc.) to the NO contacts between terminals 9 and 10 or 12 and 13 in order to activate these devices on alarm.



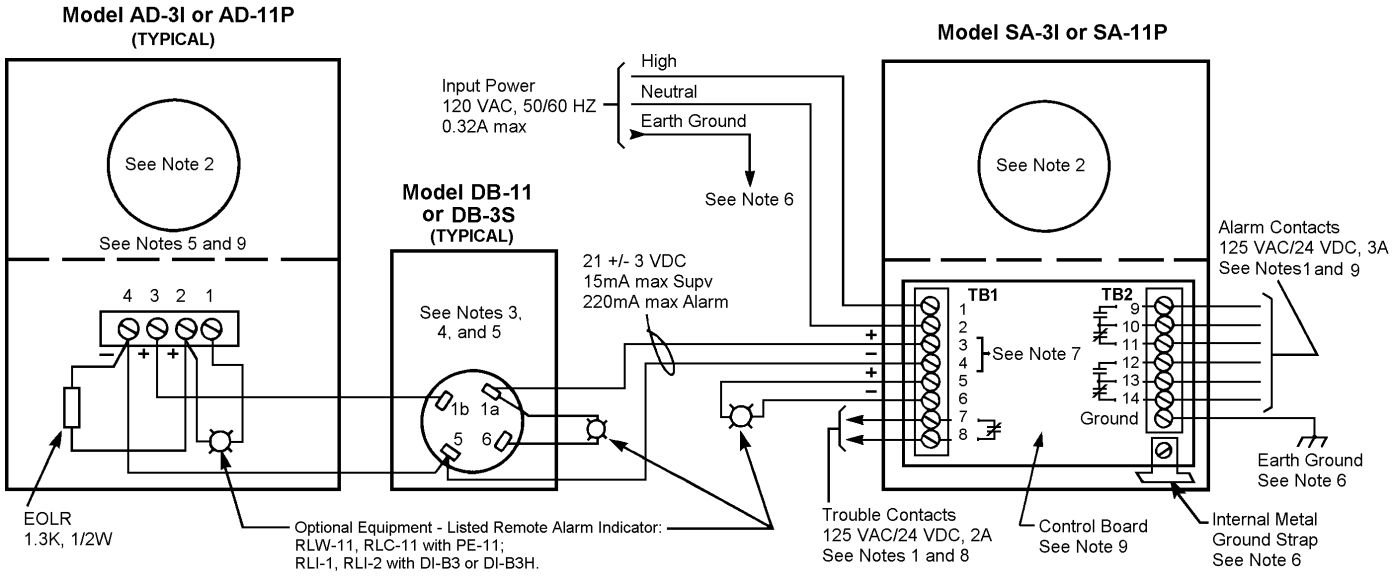
NOTE:

1. All relay contacts are shown with system in the normal supervisory, no alarm, condition.

WARNING:

If the alarm relay contacts on the control board are being used to control 120 VAC operated equipment, then ensure that earth ground is properly connected to the internal ground strap, using the proper conduit lock nuts. Failure to provide proper grounding may result in fatal electrical shock.

Figure 5
Connections for Control of Normally Energized and De-Energized Devices



CAUTION: Do not use looped wire under terminal 5 of the DB-3S or DB-11 base. Break wire run to provide supervision of both connections.

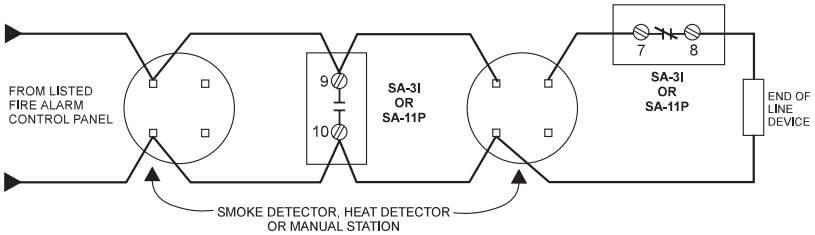
NOTES:

1. Relay contacts shown with power applied and system in normal supervisory, no alarm, condition.
2. See **Technical Data** for compatible detectors.
3. Arrangement for photo detector and base located within air duct.
4. Use PE-11 with either DB-3S base and DB-ADPT, or with DB-11 base.
5. No more than two external detectors may be connected to terminals 3 and 4 of TB1 of the SA-3I or SA-11P.
6. Ensure that the incoming earth ground wire is properly connected to the TB2 terminal marked **GND**. If using an earth grounded conduit, ensure that it is properly connected to the internal metal ground strap, using the proper conduit locking nuts. **Failure to provide proper grounding may result in fatal electrical shock.**
7. Connect EOL device (1.3K, 1/2W resistor) to terminals 3 and 4 of TB1 when no external detectors are connected.
8. When the relay contacts are connected to a fire alarm system, the trouble contacts must be wired beyond the last detector so as not to interfere with an alarm, as shown in Figure 7.
9. Use SA-3I or SA-11P relay for alarm signaling. Do not connect relay modules directly to detectors.

Figure 6
Typical Connections for Models SA-3I and SA-11P

Typical Fire Alarm System Interface

The Model SA-3I/-11P can be wired to interface to a fire alarm system as shown in Figure 7. The connections to the alarm relay (NO) contacts can be made between terminals 9 and 10 or 12 and 13. The connections to the trouble relay (NC) contacts can be made between terminals 7 and 8.



NOTES:

1. All relay contacts are shown with the SA-3I/-11P powered by 120 VAC and in the normal supervisory, no alarm, condition.
2. When connected into a fire alarm system, trouble contacts must be wired in beyond the last detector so as not to interfere with an alarm.

CAUTION: Do not use looped wire at the relay contact connections (NC, C, NO). Break wire runs at the connections to provide supervision of the connections.

Figure 7
Typical Fire Alarm System Interface

INSTALLATION OF SMOKE DETECTORS

CAUTION: Air Duct Detector Models SA-3I/-11P are designed for use with Smoke Detector Models DI-B3, DI-B3H, or PE-11. Do not use any other detector (See Figure 3, Items 14 and 15).

For DI-B3 and DI-B3H Detectors:

Installation

Align the detector alarm LED with the LED imprint located within the detector compartment on the housing. Rotate the detector counterclockwise while pushing on the detector until it drops in place. Next, push and rotate clockwise to engage the electrical connections. The detector automatically stops and locks in place.

Removal

To remove the detector, push on the detector and rotate it counterclockwise until the stop is reached. Then pull the detector out of the base to disengage.

For PE-11 Detector:

Installation of the DB-ADPT Base Adapter

a. Install the DB-ADPT adapter into the detector compartment on the housing.

b. Push the adapter into the base and rotate clockwise to engage the electrical connections. The adapter will automatically stop and lock into place.

Installation/Removal of PE-11 Detector in Base Adapter

To Install:

- Align detector with base and insert detector.
- Rotate detector counterclockwise while gently pressing on it until the detector drops fully into base.
- Then rotate the detector clockwise until it stops and locks in place. Insert optional locking screw (Order Model LK-11).

To Remove:

- Rotate the detector counterclockwise until stop is reached.
- Pull detector out of base.

Sensing Compartment Cover Installation

a. Install the cover gasket (Figure 3, 16) on the housing, if not already installed. Be certain the alignment holes in the cover gasket are aligned with the guides on the housing.

b. Install the sensing compartment ionization cover (20) when using DI-B3 or DI-B3H (15) detectors and the PE-11 cover (17) when using an PE-11 detector (14).

c. Fasten the cover, using the No. 6 screws (18) provided. Tighten snugly, but not so tightly as to warp cover and cause air leakage around the edges of the cover.

Sensitivity Testing

Test at least as frequently as required by the ***Installation and Test Requirements of NFPA 72***.

a. Siemens Industry, Inc. smoke detector sensitivities are factory set and are **not** field adjustable. The sensitivity range, as measured with the appropriate sensitivity meter, is listed on the nameplate label on the wiring compartment cover.

b. To ensure that the sensitivities are within factory-specified limits, sensitivity testing of the smoke detectors should be accomplished by qualified service personnel only.

c. The SA-3I provides access to the detector sensitivity test jack through the sensing compartment cover. Access the test jack by lifting up the hinged lid (Figure 3, 21) and inserting the plug. For further information on testing the DI-B3 and DI-B3H detectors, refer to the *TM-I3 Instructions,* P/N 315-086641.

d. The PE-11 detector monitors smoke sensitivity automatically and requires no test equipment. A green flash of the detector LED about every seven seconds indicates that the smoke sensitivity is within it listed limits.

MAINTENANCE

The performance of the air duct detector unit may be adversely affected by dirt or foreign matter on the sampling tubes or detector. If the air holes in the inlet sampling tube become restricted, the unit cannot receive a proper air sample, and performance is impaired. It is recommended that the sampling tubes be checked and cleaned periodically. The detector maintenance program should consist of periodic cleaning of dust from the detector head by a light brushing of all open slots and by using a vacuum cleaner. For cleaning of Model PE-11, refer to the Installation Instructions, P/N 315-094198.

The cleaning and test program is recommended for six month intervals, or more frequently, if needed, depending on the individual detector environment.

CAUTION: Under no circumstances is the ionization or photoelectric detector portion of the unit to be disassembled by anyone other than an authorized Siemens Industry, Inc. Systems Technician. For service, contact your nearest authorized Siemens Industry, Inc. Service Representative.

Periodic Testing

It is recommended that the unit be tested periodically to insure optimum performance. This can be done by removing the sealing plug from the inlet sampling tube on the opposite side of the duct and blowing smoke directly into the inlet tube. Check that the fire alarm system is activated.

CAUTION: If the fire alarm system is connected to a central station or fire department, or has external devices such as fans, extinguishers, etc., connected, notify appropriate personnel and disconnect the external devices until all tests are completed. After testing, reset the system by using the Reset key, reconnect the devices and notify the personnel that the system is again operating.